The nuclide $^{252}_{98}$Cf emits neutrons through spontaneous fission in ~3% of all decays, the rest being α-decays. All the other neutron sources listed involve a radioactive nuclide whose decay causes a nuclear reaction in a secondary substance which produces neutrons. For example, $^{124}_{51}$Sb produces neutrons in beryllium powder or metal as a result of the initial emission of γ-rays, in which case there is no coulomb barrier to penetrate. Radium, polonium, plutonium, and americium produce neutrons by nuclear reactions induced in beryllium by the α-particles from their radioactive decay. For the neutrons produced either by spontaneous fission in californium or by the α-particle reaction with beryllium, the energy is between 0 and 10 MeV, while for the neutrons emitted in the γ,n-reaction involving $^{124}_{51}$Sb, the energy is 0.02 MeV. Neutron sources are commercially available.

Neutron multipliers have also come into use. These usually consist of a tank containing highly enriched ^{235}U. The uranium releases 100 n for each trigger neutron sent in; the common source is ^{252}Cf. These units are used for neutron radiography (§7.9.2), activation analysis (§18.3.3), etc.

Although the neutron flux from radioactive sources is comparatively small, they are still quite useful for specific purposes, due to their extreme simplicity, reliability (they cannot be shut off), and small size. The γ-radiation which is also present is, however, a disadvantage. From this point of view, ^{252}Cf is the most preferable neutron source, but, on the other hand, it has a relatively short half-life compared with some of the other sources. ^{252}Cf is further discussed in Chapter 14.

10.10. EXERCISES

10.1. (a) What kinetic energy must be given to a helium atom in order to increase its mass by 1%? (b) What are the mean velocity and the mean kinetic energy of a helium atom at STP?

10.2. Calculate the distance of closest approach for 5 MeV α-particles to a gold target.

10.3. In a Rutherford scattering experiment ^{2}H atoms of 150 keV are used to bombard a thin ^{58}Ni foil having a surface density of 67×10^{-6} g cm^{-2}. The detector subtends a solid angle of 1.12×10^{-4} sr and detects 4816 deuterons out of a total of 1.88×10^{12} incident on target. Calculate (a) the differential cross-section (in barns). (b) What is the distance between target and the solid state detector, which has a surface area of 0.2 cm^2?

10.4. In Rutherford scattering on a silver foil using α-particles from a thin-walled radon tube, the following data were observed: $d\sigma/d\Omega = 22(\theta\ 150°)$, 47(105°), 320(60°), 5260(30°), 105 400(15°) barns per steradian. Calculate the energy of the incident α-particles.

10.5. Alpha-particles from ^{218}Po (E_α 6.0 MeV) are used to bombard a gold foil. (a) How close to the gold nucleus can these particles reach? (b) What is the nuclear radius of gold according to the radius-mass relation ($r_o = 1.3$ fm)?

10.6. What is the *Q*-value for the reactions: (a) $^{11}B(d,\alpha)^{9}Be$; (b) $^{7}Li(p,n)^{7}Be$?

10.7. What is the maximum velocity that a deuteron of 2 MeV can impart to a ^{16}O atom?

10.8. Calculate the mass of an electron accelerated through a potential of 2×10^{8} V.

10.9. ^{12}C atoms are used to irradiate ^{239}Pu to produce an isotope of berkelium. What is the Coulomb barrier height?

10.10. Measurements made on the products of the reaction $^{7}Li(d,\alpha)^{5}He$ have led to an isotopic mass of 5.0122 for the hypothetical nuclide ^{5}He. Show that this nuclear configuration cannot be stable by considering the reaction $^{5}He \rightarrow {}^{4}He + n$.

10.11. In an experiment one hopes to produce the long-lived (2.6 y) ^{22}Na through a d,2n-reaction on neon. What is (a) the *Q*-value, (b) the threshold energy, (c) the Coulomb barrier height, and (d) the minimum deuteron energy for the reaction? The mass excesses (in keV) are −5185 for ^{22}Na and −8027 for ^{22}Ne.

10.11. LITERATURE

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